Top8000 Dataset — General Statistics and Geometric Analysis

Residues: 1,568,561 | **Chains:** 7,814 | **Structures:** 7,607 | **Source:** top8000_measures.jsonl | **Generated by:** pydangle-biopython v0.5.1

This document provides the following analyses:

1. **General Information** — dataset overview, measurements available, and summary statistics
2. **Amino Acid Composition and Chain Length Distribution** — comparison with UniProt and PDB reference databases
3. **Torsion Angle Distributions** — univariate (backbone, omega deviation, sidechain) and multivariate (Ramachandran) distributions using circular statistics
4. **Bond Angle Distributions** — tau (N–Cα–C) and other non-periodic angles using linear statistics
5. **References**

1. General Information

Dataset	Top8000
Total residues	1,568,561
Unique structures	7,607
Unique chains	7,814

Measurements available: phi: 1,523,961, psi: 1,523,951, omega: 1,523,961, tau: 1,568,552, chi1: 1,309,084, chi2: 991,054, chi3: 595,690, chi4: 283,007. Not all residues have all measurements — terminal residues lack phi/psi/omega, and glycine and alanine lack sidechain chi angles.

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Ramachandran category distribution:
  General      1,056,115  (67.33%)
  IleVal        200,462  (12.78%)
  Gly           112,610  ( 7.18%)
  TransPro       63,261  ( 4.03%)
  PrePro         50,230  ( 3.20%)
  CisPro          3,524  ( 0.22%)

Torsion angles (circular statistics):
  Angle             N    Circ Mean     Circ Std
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  phi       1,523,961      -85.131       41.641
  psi       1,523,951       10.017      115.055
  omega     1,523,961      179.273        7.902
  chi1      1,309,084      -88.407       85.012


  chi2        991,054      167.494      110.466
  chi3        595,690     -173.251      102.960
  chi4        283,007       -3.321      105.364

Bond angles (linear statistics):
  tau       1,568,552      111.121        2.715
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2. Amino Acid Composition and Chain Length Distribution

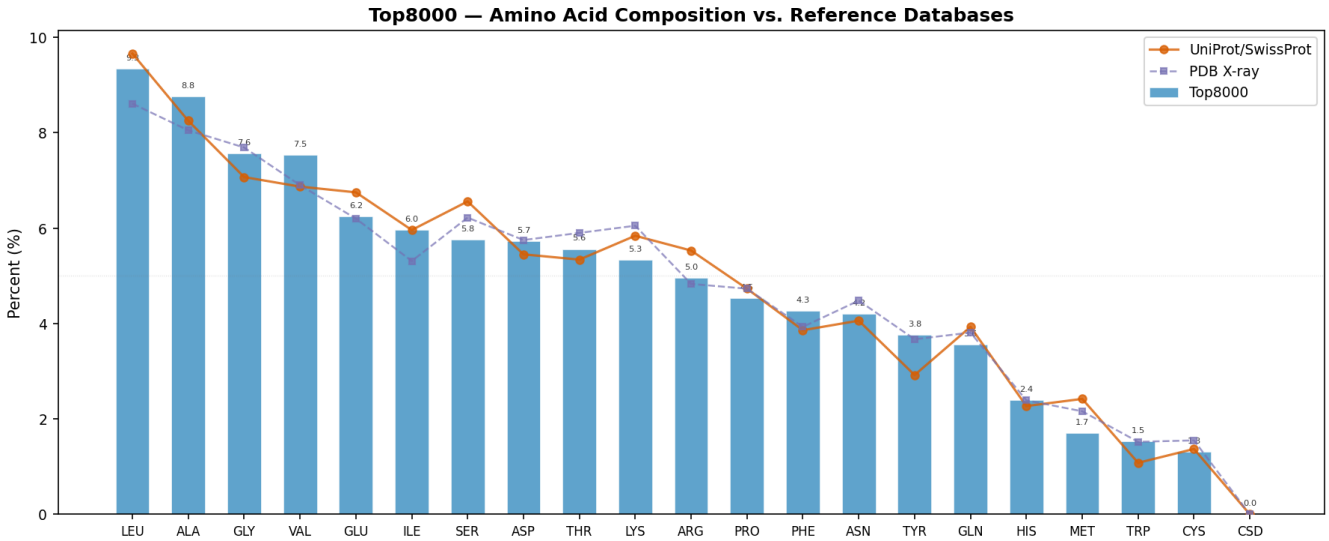
2.1 Representativeness

The figure and table below compare the Top8000 amino acid frequencies against two reference databases:

- **UniProt/SwissProt** (~2024): amino acid frequencies across all reviewed protein sequences, representing the broadest available view of protein sequence space.
- **PDB X-ray** (~2025): frequencies from a 10,000-entity sample of X-ray crystallographic structures in the RCSB Protein Data Bank.

The two reference distributions are highly similar (most amino acids differ by <0.5%), indicating that the PDB's well-known crystallization bias — overrepresentation of soluble, globular, well-expressing proteins from model organisms — has only a modest effect on overall amino acid composition.

The Top8000 dataset (1,568,561 residues from 7,607 structures) shows additional deviations from both references due to quality filtering, homology reduction, and sample size effects.



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Residue	Count	Dataset%	UniProt%	PDB%	Δ UniProt
LEU	146,611	9.35%	9.66%	8.61%	-0.31%
ALA	137,293	8.75%	8.25%	8.05%	+0.50%
GLY	118,710	7.57%	7.07%	7.69%	+0.50%
VAL	118,127	7.53%	6.87%	6.90%	+0.66%
GLU	97,991	6.25%	6.75%	6.20%	-0.50%
ILE	93,569	5.97%	5.96%	5.31%	+0.01%
SER	90,238	5.75%	6.56%	6.22%	-0.81%
ASP	89,811	5.73%	5.45%	5.75%	+0.28%
THR	87,232	5.56%	5.34%	5.90%	+0.22%
LYS	83,548	5.33%	5.84%	6.05%	-0.51%
ARG	77,862	4.96%	5.53%	4.83%	-0.57%

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PRO        71,026     4.53%     4.73%     4.73%     -0.20%
PHE        66,832     4.26%     3.86%     3.93%     +0.40%
ASN        65,830     4.20%     4.06%     4.48%     +0.14%
TYR        58,959     3.76%     2.92%     3.67%     +0.84%
GLN        55,838     3.56%     3.93%     3.81%     -0.37%
HIS        37,563     2.39%     2.27%     2.39%     +0.12%
MET        26,804     1.71%     2.42%     2.16%     -0.71%
TRP        24,087     1.54%     1.08%     1.52%     +0.46%
CYS        20,629     1.32%     1.37%     1.55%     -0.05%
CSD             1     0.00%     0.00%     0.00%     +0.00%
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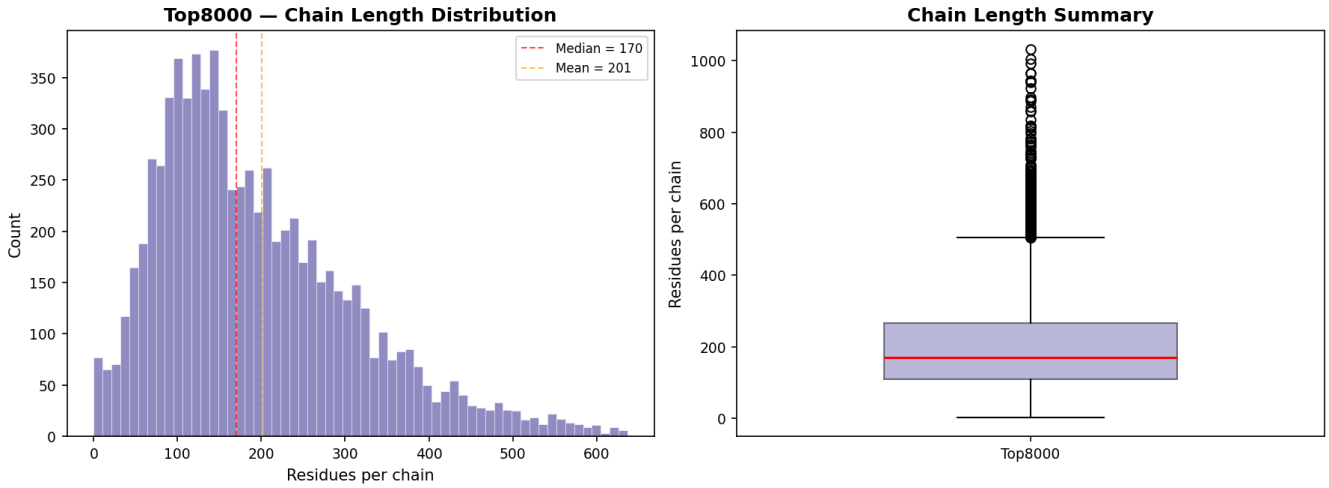
2.2 Chain Length Distribution



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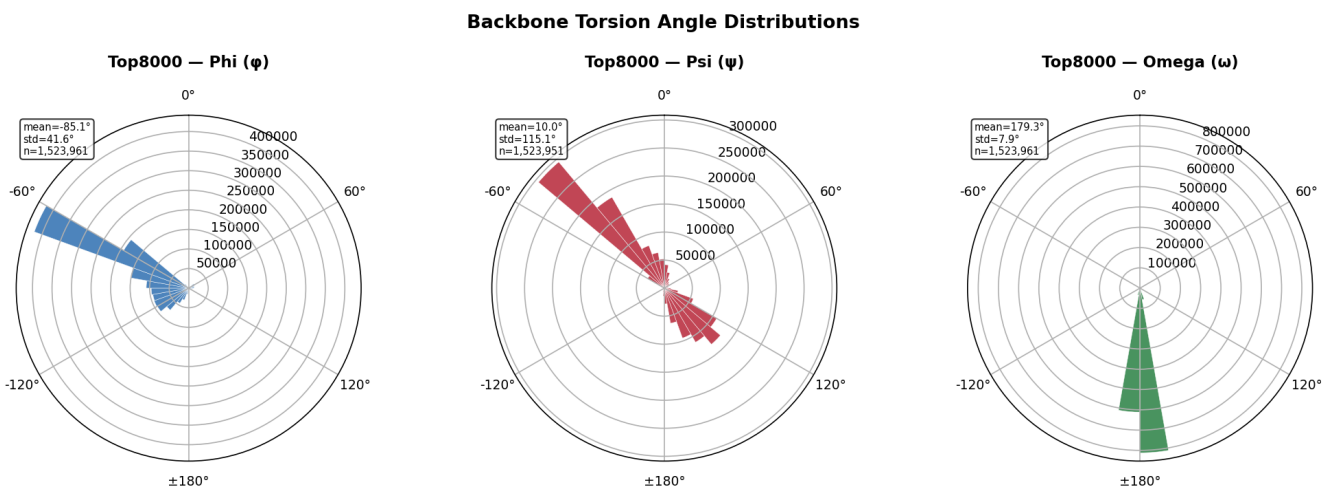
Chain length statistics (7,814 chains):
  Min:             1
  Q1:            108
  Median:        170
  Mean:        200.7
  Q3:            267
  Max:         1,033
  Std:         130.8
```

3. Torsion Angle Distributions (Circular Statistics)

Torsion angles (φ, ψ, ω, χ) are periodic variables defined on the interval [–180°, +180°]. Standard (linear) arithmetic means and standard deviations give misleading results when values cluster near the ±180° boundary — for example, a linear mean of trans peptide bond ω values near +179° and –179° yields ~0° rather than the correct ~180°. All torsion angle summary statistics in this report use **circular statistics**: the circular mean is computed as atan2(⟨sin θ⟩, ⟨cos θ⟩), and the circular standard deviation as √(–2 ln R) where R is the mean resultant length.

3.1 Univariate Distributions

3.1.1 Backbone Torsion Angles (φ, ψ, ω)



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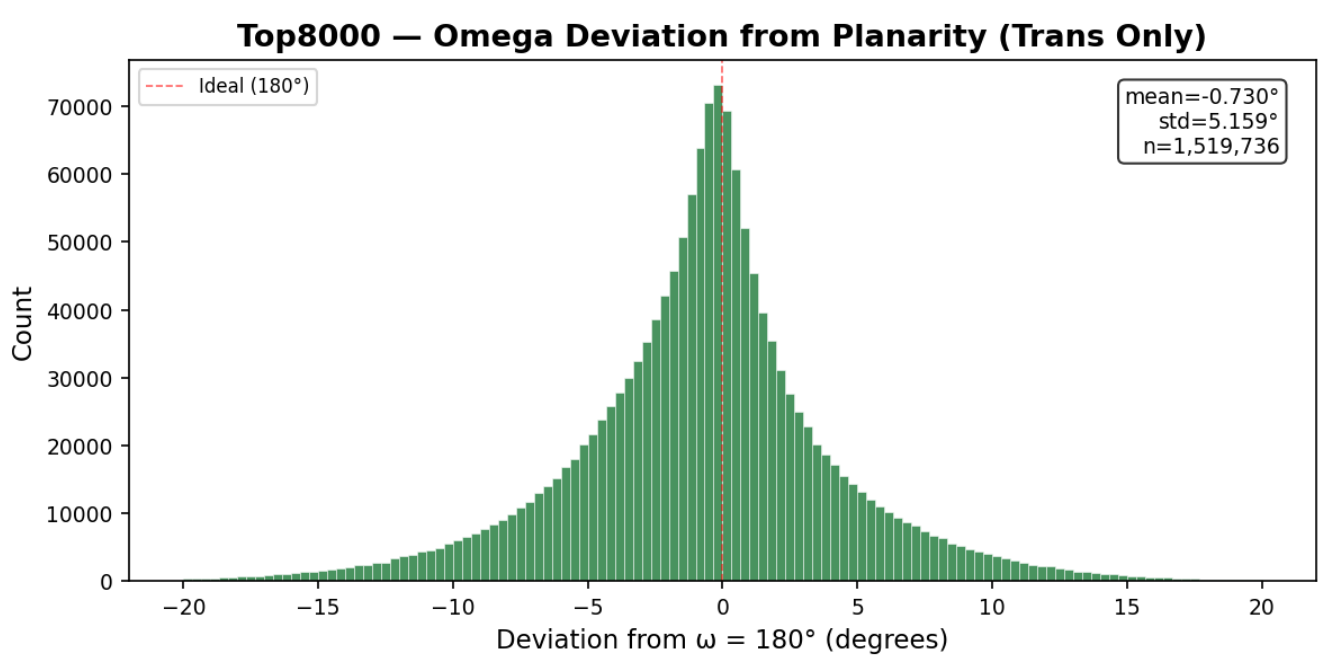
3.1.2 Omega Deviation from Planarity (Trans Peptides Only)



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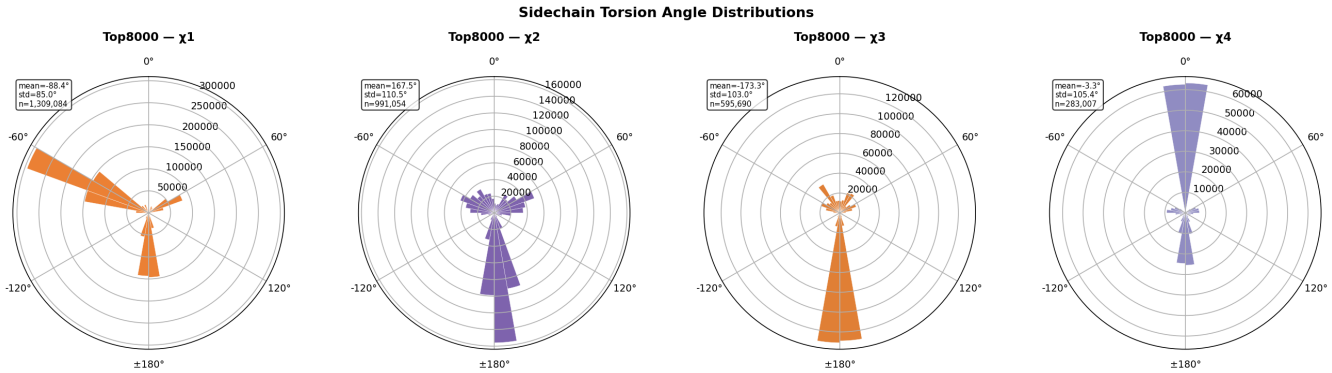
3.1.3 Sidechain Torsion Angles (χ1–χ4)



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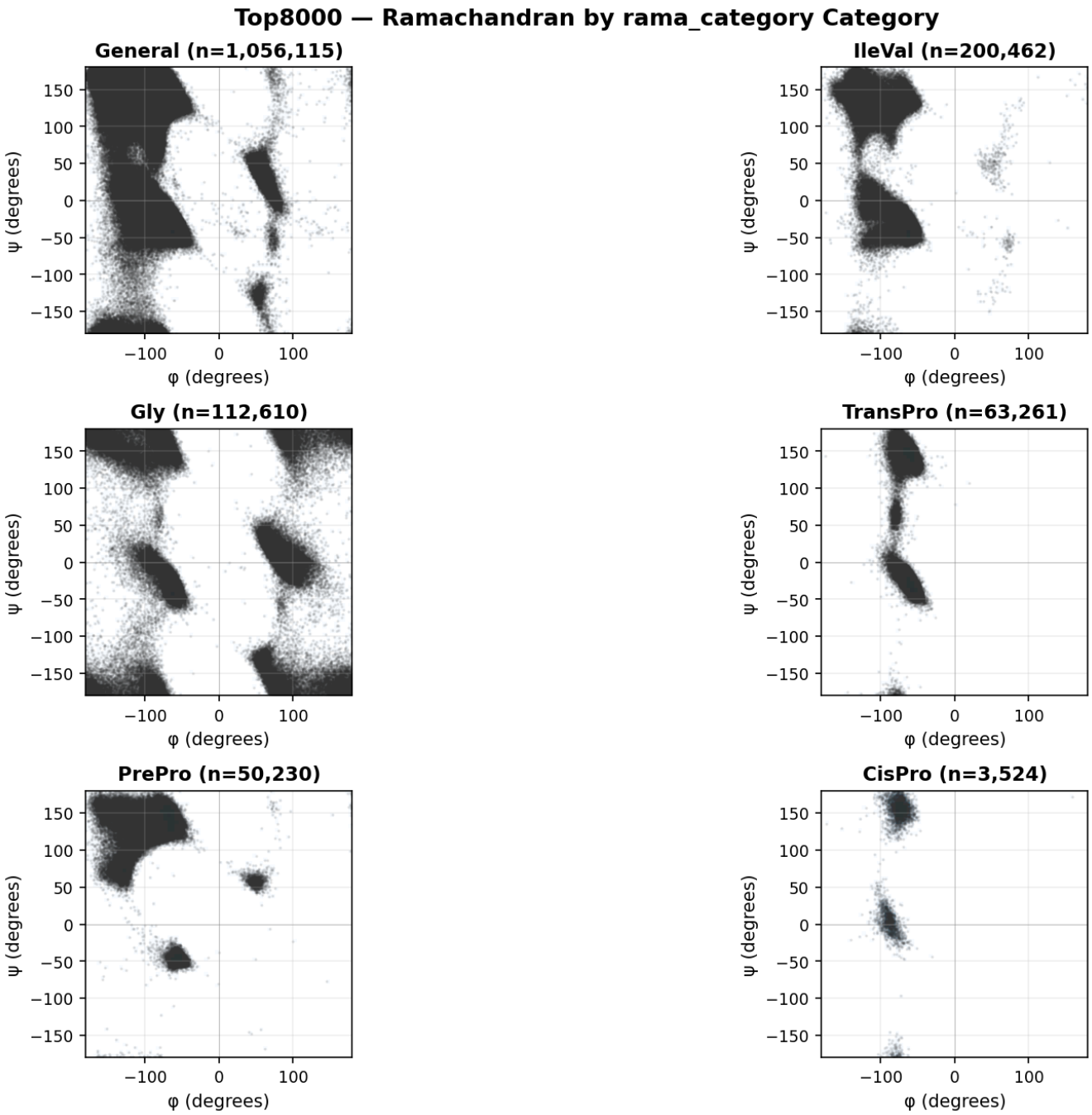
3.2 Multivariate Distributions

3.2.1 Ramachandran Distributions (φ × ψ by Category)



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Category counts (rama_category):
  General      1,056,115  (67.33%)
  IleVal         200,462  (12.78%)
  Gly            112,610  ( 7.18%)
  TransPro        63,261  ( 4.03%)
  PrePro          50,230  ( 3.20%)
  CisPro           3,524  ( 0.22%)
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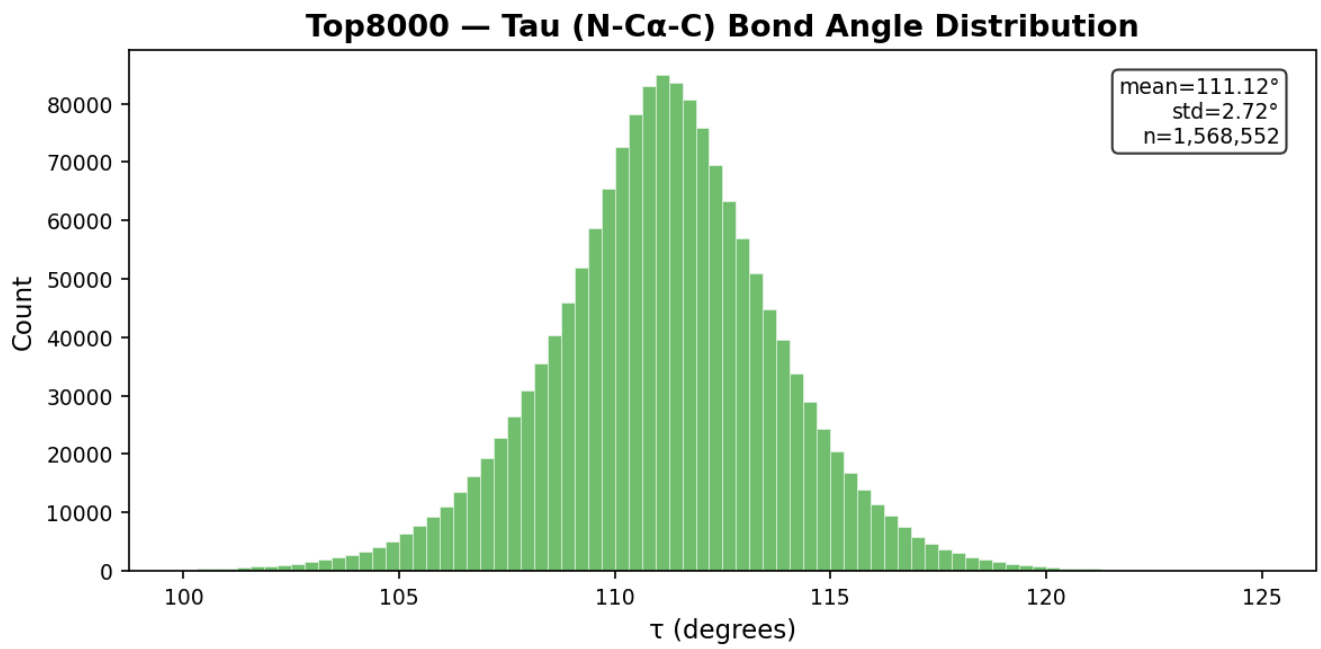
4. Bond Angle Distributions (Linear Statistics)

Bond angles are non-periodic and are analyzed with standard linear statistics (arithmetic mean, standard deviation).

4.1 Tau (N–Cα–C)



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5. References

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